

31. To find the greatest possible S.P. of the most expensive product, we need to consider the minimum S.P. of the remaining 24 products which is ₹ 420 each for 10 products and ₹ 1000 each for other 14 products.

$$\text{Minimum S.P. of 24 products} = ₹ (420 \times 10 + 1000 \times 14) = ₹ (4200 + 14000) = ₹ 18200.$$

$$\text{Total S.P. of 25 products} = ₹ (1200 \times 25) = ₹ 30000.$$

$$\therefore \text{Greatest possible S.P. of the most expensive product} = ₹ (30000 - 18200) = ₹ 11800.$$

32. Number of cars sold per week originally = $15 \times 7 = 105$.
Number of cars sold per week under scheme = $21 \times 7 = 147$.
Increase in sale = $147 - 105 = 42$.

$$\therefore \text{Percentage increase} = \left(\frac{42}{105} \times 100 \right) \% = 40\%.$$

$$33. 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

$$\therefore 1^2 + 2^2 + 3^2 + \dots + 7^2 = \left(\frac{7 \times 8 \times 15}{6} \right) = 140.$$

$$\text{So, required average} = \left(\frac{140}{7} \right) = 20.$$

34. Clearly, $b = a + 2$, $c = a + 4$, $d = a + 6$ and $e = a + 8$.

$$\therefore \text{Average} = \frac{a + (a+2) + (a+4) + (a+6) + (a+8)}{5} = \left(\frac{5a+20}{5} \right) = (a+4).$$

35. Let the number be x . Then,

$$\begin{aligned} \frac{x+x^2}{2} &= 5x \Leftrightarrow x^2 - 9x = 0 \\ &\Leftrightarrow x(x-9) = 0 \\ &\Leftrightarrow x = 0 \text{ or } x = 9. \end{aligned}$$

So, the number is 9.

36. $S + M + Sc = 100 \times 3 = 300$ and $S + M = Sc + 80$

$$\Rightarrow Sc + 80 + Sc = 300 \Rightarrow 2Sc = 220 \Rightarrow Sc = 110.$$

37. The given series is a G.P. with first term, $a = 1$ and common ratio, $r = 2$. It has $(n+1)$ terms.

$$\therefore \text{Sum of the terms of the series} = \frac{(2^{n+1} - 1)}{(2 - 1)} = 2^{n+1} - 1.$$

$$\text{Arithmetic mean} = \frac{2^{n+1} - 1}{n+1}.$$

38. Let the two numbers be x and y .

$$\text{Then, } x + y = 6.5 \times 2 = 13 \text{ and } \sqrt{xy} = 6 \text{ or } xy = 36.$$

$$(x-y)^2 = (x+y)^2 - 4xy = (13)^2 - 4 \times 36 = 169 - 144 = 25 \\ \Rightarrow (x-y) = 5.$$

$$\text{Solving } x + y = 13 \text{ and } x - y = 5, \text{ we get: } x = 9, y = 4.$$

39. Let the four numbers be a , b , c and d respectively.

$$\text{Then, } a = \frac{1}{4}(b+c+d) \Rightarrow b+c+d = 4a.$$

$$\text{Also, } a+b+c+d = 60 \times 4 = 240$$

$$\Rightarrow a+4a = 240 \Rightarrow 5a = 240 \Rightarrow a = 48.$$

Hence, first number = 48.

40. Sum of four integers = $59 \times 4 = 236$.

Let the required integers be x and $x - 28$.

$$\text{Then, } x + (x - 28) = 236 - (83 + 29) = 124$$

$$\Rightarrow 2x = 152 \Rightarrow x = 76.$$

Hence, required integer = 76.

41. Let the numbers be x , $x + 1$, $x + 2$, $x + 3$, $x + 4$, $x + 5$ and $x + 6$.

Then,

$$\frac{x + (x+1) + (x+2) + (x+3) + (x+4) + (x+5) + (x+6)}{7} = 20$$

$$\text{or } 7x + 21 = 140 \text{ or } 7x = 119 \text{ or } x = 17.$$

$$\therefore \text{Largest number} = x + 6 = 23.$$

42. Let the numbers be x , $x + 2$, $x + 4$, $x + 6$ and $x + 8$.

$$\text{Then, } \frac{x + (x+2) + (x+4) + (x+6) + (x+8)}{5} = 95 \text{ or } 5x + 20$$

$$= 475 \text{ or } 5x = 455 \text{ or } x = 91.$$

So, the numbers are 91, 93, 95, 97 and 99.

Clearly, the fourth number in the descending order is 93.

43. Let x , $x + 2$, $x + 4$ and $x + 6$ represent numbers A , B , C and D respectively.

$$\text{Then, } \frac{x + (x+2) + (x+4) + (x+6)}{4} = 65$$

$$\Rightarrow 4x + 12 = 260$$

$$\Rightarrow 4x = 248 \Rightarrow x = 62.$$

$$\text{So, } A = 62, B = 64, C = 66, D = 68.$$

$$\therefore A \times D = 62 \times 68 = 4216.$$

44. Let the numbers be x , $x + 2$ and $x + 4$.

$$\text{Then, } (x + x + 2 + x + 4) - \frac{(x + x + 2 + x + 4)}{3} = 44$$

$$\Rightarrow (3x + 6) - \frac{(3x + 6)}{3} = 44$$

$$\Rightarrow 2(3x + 6) = 132$$

$$\Rightarrow 6x = 120 \Rightarrow x = 20.$$

$$\therefore \text{Largest number} = x + 4 = 24.$$

45. Sum of marks in 4 subjects = $75 \times 4 = 300$.

$$\text{Sum of marks in 5 subjects} = 300 + 80 = 380.$$

$$\therefore \text{New average} = \frac{380}{5} = 76.$$

46. Clearly, to find the average, we ought to know the number of boys, girls or students in the class, neither of which has been given.

So, the data provided is inadequate.

47. Let the number of other workers be x . Then, number of agricultural workers = $11x$.

$$\text{Total number of workers} = 12x.$$

$$\therefore \text{Average monthly income} = \frac{S \times 11x + T \times x}{12x} = \frac{11S + T}{12}.$$

$$48. \text{Required average} = \left(\frac{67 \times 2 + 35 \times 2 + 6 \times 3}{2 + 2 + 3} \right)$$

$$= \left(\frac{134 + 70 + 18}{7} \right) = \frac{222}{7} = 31\frac{5}{7} \text{ years.}$$